

Create a Spring Web Project using Maven

The screenshot displays the Eclipse IDE interface for a Spring Boot project named 'spring-learn'. The project structure in the left sidebar includes 'src/main/java' containing 'com.cognizant.spring_learn' and 'SpringLearnApplication.java', 'src/main/resources', 'src/test/java', 'JRE System Library [JavaSE-17]', 'Maven Dependencies', 'src', 'target', 'HELP.md', 'mvnw', 'mvnw.cmd', and 'pom.xml'. The main editor shows the code for 'SpringLearnApplication.java' with the following content:

```
1 package com.cognizant.spring_learn;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class SpringLearnApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(SpringLearnApplication.class, args);
11     }
12 }
13
14
```

The right sidebar shows the 'Outline' view with the package 'com.cognizant.spring_learn' and the class 'SpringLearnApplication' containing the 'main(String[]) : void' method. The bottom console shows the output of running the application:

```
SpringLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14-hotspot\bin\javaw.exe (02-Jul-2026, 10:16:29 pm elapsed)

Starting SpringLearnApplication using Java 17.0.14 with PID 30620 (C:\Users\USER\Downloads\eclipse-workspace\spring-learn\sp
No active profile set, falling back to 1 default profile: "default"
Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
Tomcat initialized with port 8080 (http)
Starting service [Tomcat]
Starting Servlet engine: [Apache Tomcat/11.0.22]
Root WebApplicationContext: initialization completed in 1938 ms
Tomcat started on port 8080 (http) with context path '/'
Started SpringLearnApplication in 4.073 seconds (process running for 4.854)
```

Spring Core – Load Country from Spring Configuration XML

The screenshot displays the Eclipse IDE interface. The main editor shows the `SpringLearnApplication.java` file, which is a Spring Boot application. The code includes imports for `ApplicationContext` and `ClassPathXmlApplicationContext`, a `@SpringBootApplication` annotation, and a `SpringLearnApplication` class with a `main` method. The `main` method logs "START", runs the application, displays the country, and logs "END".

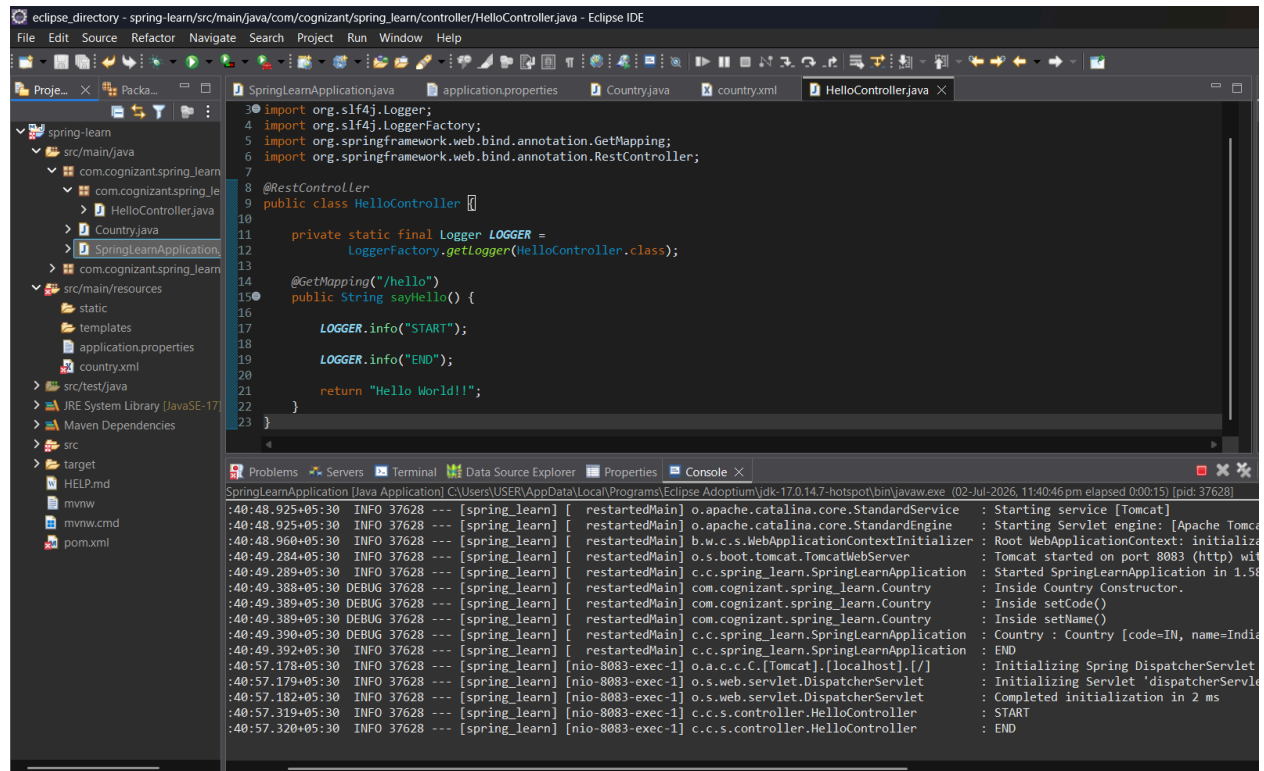
The left sidebar shows the project structure for `spring-learn`, including `src/main/java` (containing `Country.java` and `SpringLearnApplication.java`) and `src/main/resources` (containing `static`, `templates`, `application.properties`, and `country.xml`).

The bottom console window shows the output of the application, indicating that it started successfully using Java 17.0.14 with Spring Boot v4.1.0. The output also shows the initialization of Tomcat and the loading of the `Country` class from the XML configuration.

```
SpringLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14-hotspot\bin\javaw.exe (02-Jul-2026, 11:12:46 pm elapsed)

n [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.14 with PID :
n [ restartedMain] c.c.spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.0, Spring v7.0.8
n [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
n [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-logging-mbeans=true' to enable the logging of devtools property defaults.
n [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.org.springframework.boot.web.servlet.context.listener' to 'WARN' or 'DEBUG'.
n [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
n [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
n [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
n [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 2063 ms
n [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
n [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 4.01 seconds (process running for 4.01 seconds)
n [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
n [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
n [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
n [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country [code=IN, name=India]
n [ restartedMain] c.c.spring_learn.SpringLearnApplication : END
```

Hello World RESTful Web Service

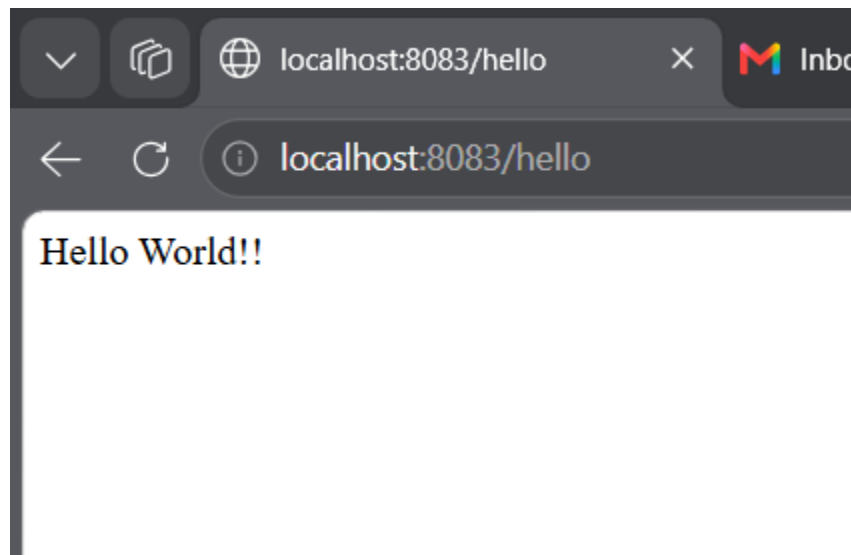


The screenshot shows the Eclipse IDE with the following components:

- Project Explorer:** Displays the project structure for 'spring-learn', including 'src/main/java/com/cognizant/spring_learn' and 'src/main/resources'.
- Editor:** Shows the source code of 'HelloController.java'. The code is as follows:

```
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.web.bind.annotation.GetMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class HelloController {
10
11     private static final Logger LOGGER =
12         LoggerFactory.getLogger(HelloController.class);
13
14     @GetMapping("/hello")
15     public String sayHello() {
16
17         LOGGER.info("START");
18
19         LOGGER.info("END");
20
21         return "Hello World!!";
22     }
23 }
```
- Console:** Displays the output of the application, showing the startup sequence of Tomcat and the successful execution of the 'sayHello' endpoint.

```
SpringLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14-hotspot\bin\javaw.exe (02-Jul-2026, 11:40:46 pm elapsed 0:00:15) [pid: 37628]
:40:48.925+05:30 INFO 37628 --- [spring_learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
:40:48.925+05:30 INFO 37628 --- [spring_learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.90]
:40:48.960+05:30 INFO 37628 --- [spring_learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization started
:40:49.284+05:30 INFO 37628 --- [spring_learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with 128 threads
:40:49.289+05:30 INFO 37628 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.58 seconds
:40:49.388+05:30 DEBUG 37628 --- [spring_learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside Country Constructor.
:40:49.389+05:30 DEBUG 37628 --- [spring_learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setCode()
:40:49.389+05:30 DEBUG 37628 --- [spring_learn] [ restartedMain] com.cognizant.spring_learn.Country : Inside setName()
:40:49.390+05:30 DEBUG 37628 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country [code=IN, name=India]
:40:49.392+05:30 INFO 37628 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : END
:40:57.178+05:30 INFO 37628 --- [spring_learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
:40:57.179+05:30 INFO 37628 --- [spring_learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
:40:57.182+05:30 INFO 37628 --- [spring_learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
:40:57.319+05:30 INFO 37628 --- [spring_learn] [nio-8083-exec-1] c.c.s.controller.HelloController : START
:40:57.320+05:30 INFO 37628 --- [spring_learn] [nio-8083-exec-1] c.c.s.controller.HelloController : END
```



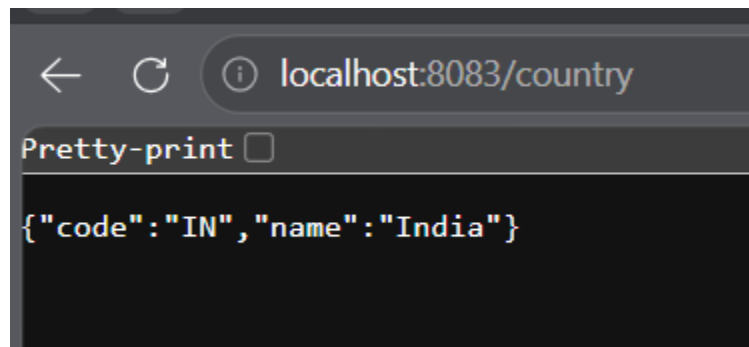
REST - Country Web Service

The screenshot shows the Eclipse IDE with the following components:

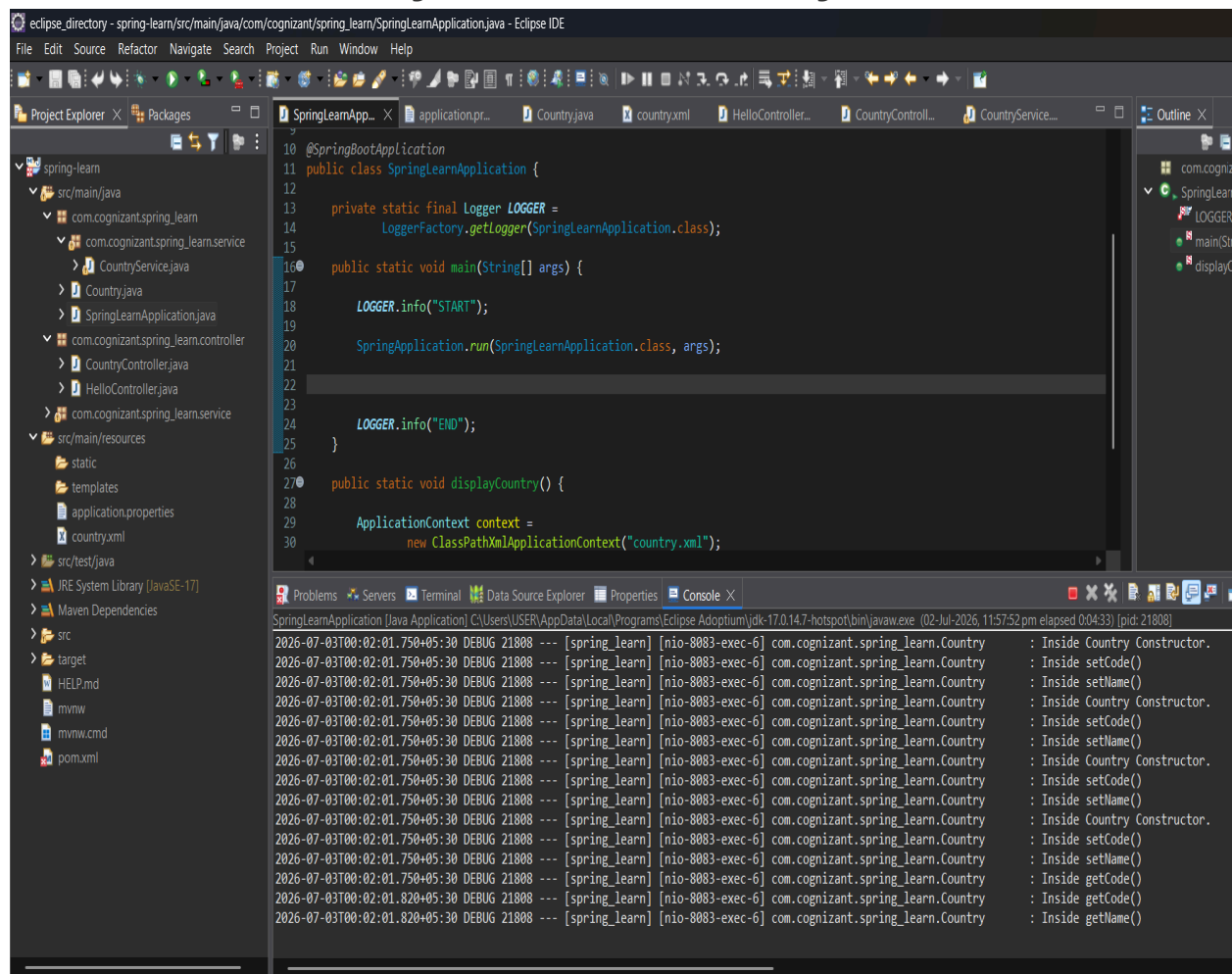
- Project Explorer:** Displays the project structure for 'spring-learn', including 'src/main/java' and 'src/main/resources'.
- Editor:** Shows the 'Country.java' file with the following code:

```
1 package com.cognizant.spring_learn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 public class Country {
7
8     private static final Logger LOGGER =
9         LoggerFactory.getLogger(Country.class);
10
11     private String code;
12     private String name;
13
14     public Country() {
15         LOGGER.debug("Inside Country Constructor.");
16     }
17
18     public String getCode() {
19         LOGGER.debug("Inside getCode()");
20         return code;
21     }
22 }
```
- Console:** Displays the application logs, showing the startup sequence and the execution of the 'Country' class.

```
SpringLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14-hotspot\bin\javaw.exe (02-Jul-2026, 11:46:25 pm elapsed 0:00:50) [pid: 1940]
2026-07-02T23:46:27.858+05:30 DEBUG 19400 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Country : Country [
2026-07-02T23:46:27.860+05:30 INFO 19400 --- [spring_learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : END
2026-07-02T23:46:53.670+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring
2026-07-02T23:46:53.672+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet
2026-07-02T23:46:53.672+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization
2026-07-02T23:46:53.738+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-1] c.c.s.controller.HelloController : START
2026-07-02T23:46:53.738+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-1] c.c.s.controller.HelloController : END
2026-07-02T23:46:58.236+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-2] c.c.s.controller.CountryController : START
2026-07-02T23:46:58.265+05:30 DEBUG 19400 --- [spring_learn] [nio-8083-exec-2] com.cognizant.spring_learn.Country : Inside Country Cons
2026-07-02T23:46:58.269+05:30 DEBUG 19400 --- [spring_learn] [nio-8083-exec-2] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-02T23:46:58.270+05:30 INFO 19400 --- [spring_learn] [nio-8083-exec-2] com.cognizant.spring_learn.Country : Inside setName()
2026-07-02T23:46:58.341+05:30 DEBUG 19400 --- [spring_learn] [nio-8083-exec-2] c.c.s.controller.CountryController : END
2026-07-02T23:46:58.341+05:30 DEBUG 19400 --- [spring_learn] [nio-8083-exec-2] com.cognizant.spring_learn.Country : Inside getCode()
2026-07-02T23:46:58.341+05:30 DEBUG 19400 --- [spring_learn] [nio-8083-exec-2] com.cognizant.spring_learn.Country : Inside getName()
```



REST - Get country based on country code

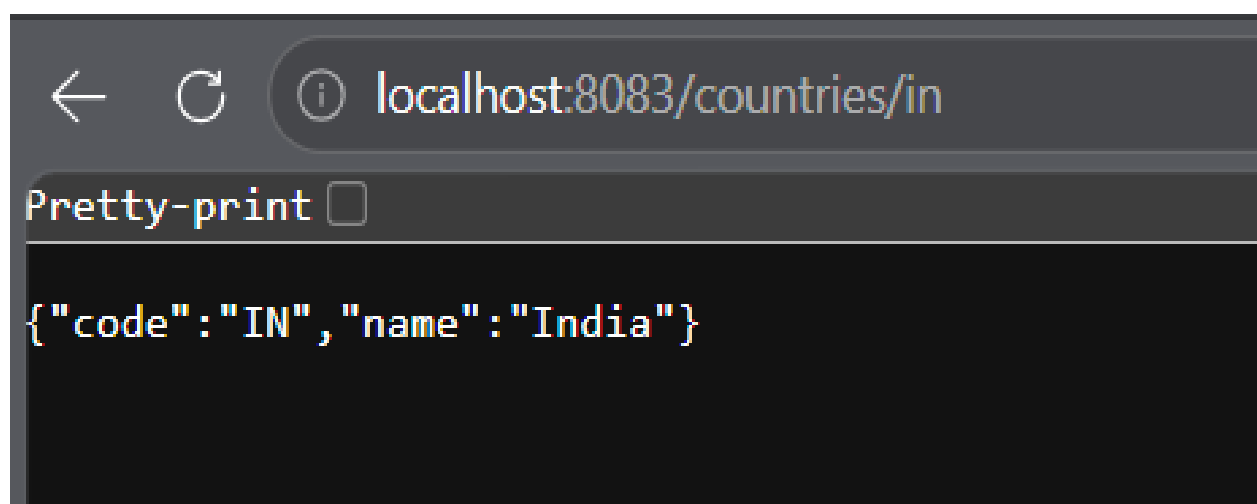


The screenshot shows the Eclipse IDE with the SpringLearnApplication.java file open. The code defines a SpringBootApplication with a main method and a displayCountry method. The console output shows the application running and logging the start and end of the process, along with the execution of the displayCountry method.

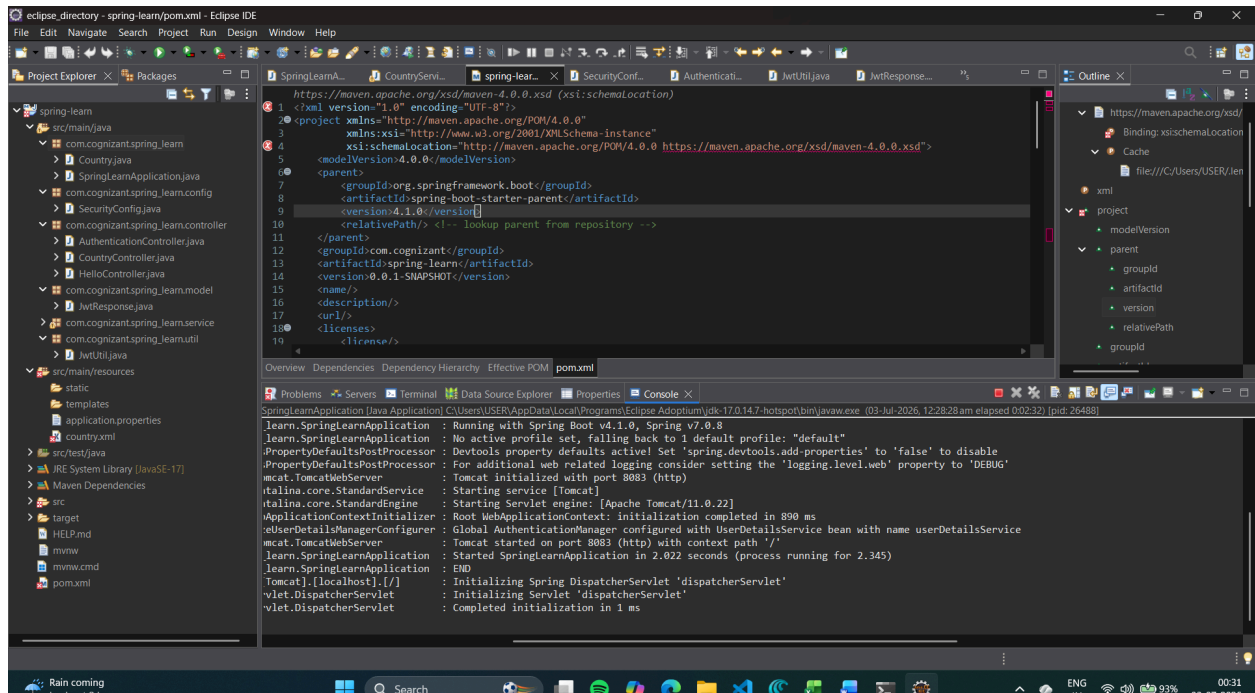
```
10 @SpringBootApplication
11 public class SpringLearnApplication {
12
13     private static final Logger LOGGER =
14         LoggerFactory.getLogger(SpringLearnApplication.class);
15
16     public static void main(String[] args) {
17
18         LOGGER.info("START");
19
20         SpringApplication.run(SpringLearnApplication.class, args);
21
22
23
24         LOGGER.info("END");
25     }
26
27     public static void displayCountry() {
28
29         ApplicationContext context =
30             new ClassPathXmlApplicationContext("country.xml");
```

Console Output:

```
SpringLearnApplication [Java Application] C:\Users\USER\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.14-hotspot\bin\java.exe (02-Jul-2026, 11:57:52 pm elapsed 0:04:33) [pid: 21808]
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside setName()
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside setName()
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside Country Constructor.
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside setCode()
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside setName()
2026-07-03T00:02:01.750+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside getCountry()
2026-07-03T00:02:01.820+05:30 DEBUG 21808 --- [spring_learn] [nio-8083-exec-6] com.cognizant.spring_learn.Country : Inside getCountry()
```



Create authentication service that returns JWT



SUCCESS: The process with PID 29184 has been terminated.

```
C:\Users\USER\Downloads\eclipse-workspace\spring-learn\spring-learn>curl -u
user:pwd http://localhost:8083/authenticate
{"token":"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJlc2VyIiwiaWF0IjoxNzgzMDE4NzQwLCJle
HAiOiJlE3ODMwMTk5NDY9LmC1_OHpfaIpKb6I5MSvHSGmyTdOgEzgm790q0nohUWQ"}
C:\Users\USER\Downloads\eclipse-workspace\spring-learn\spring-learn>curl -u
user:pwd http://localhost:8083/authenticate
{"token":"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJlc2VyIiwiaWF0IjoxNzgzMDE4ODEwLCJle
HAiOiJlE3ODMwMTk5NDY9LmC1_OHpfaIpKb6I5MSvHSGmyTdOgEzgm790q0nohUWQ"}
C:\Users\USER\Downloads\eclipse-workspace\spring-learn\spring-learn>
```